AI Vision Engineer — Task Assignment

1. **Objective**

Build an AI Vision solution that detects **lane violation behavior** from traffic video.

The core problem is based on the rule that a vehicle must be driven entirely within a single lane. If a tire touches or overlaps a lane boundary, or a vehicle moves across lane boundaries, that behavior may be relevant for lane-discipline analysis.

In practice, detecting and proving a legal violation precisely can be difficult and requires many assumptions. Since you have limited time, you are not required to make a final legal conclusion. Your task is to detect the visual events that could later support such analysis.

2. **Problem Statement**

All the example above from the video:

https://www.youtube.com/watch?v=eO19UTm93GQ

Your system should process a traffic video and detect the following:

1. Lane and Line Detection

Your system should detect lane boundary lines and lane separation from the video. Label each detected lane and line with an ID, starting from **1**

If the road has 4 detected lane lines, then the system should infer 3 lanes.

This lane and line detection result should be shown as an image output for easy viewing in the UI.

2. Vehicle Touching a Lane Line

Detect cases where a vehicle touches a lane boundary line.

Example:

- 00:06 - 00:07 VE586 grey car touch line

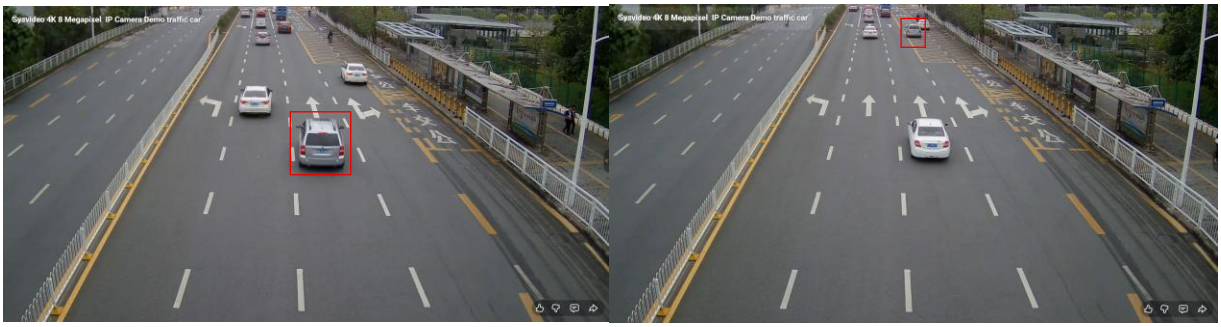
The output video should highlight the vehicle (vehicle_id) and show the event description.

3. Vehicle Changing Lane / Crossing a Lane

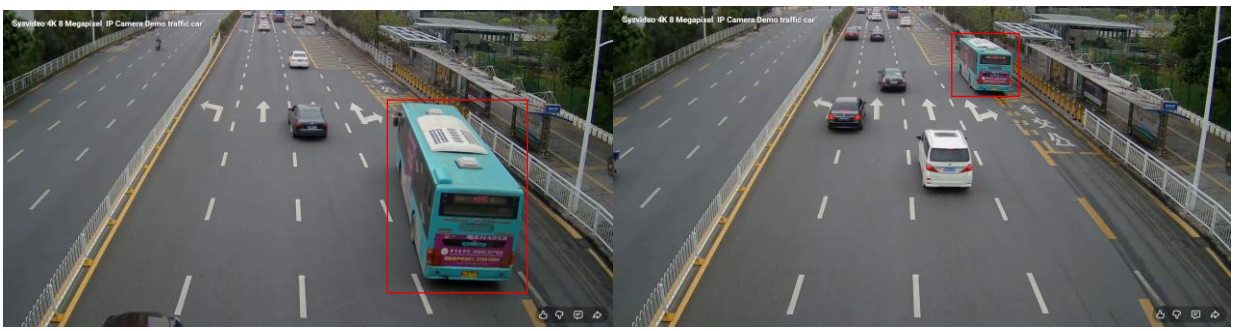
Detect cases where a vehicle moves from one lane to another or crosses a lane boundary over a period of time.

Example:

- 00:08 - 00:11 VE586 grey car change lane



- 00:14 - 00:17 N4010 blue bus change lane



The output video should highlight the vehicle (vehicle_id) and show the event description.

4. Vehicle Turn Signal Request for Lane Change

Detect when a vehicle turns on a lane-change signal and identify which signal it is using (left or right)

Example:

- 00:25 - 00:40 97963 black car ask for right signal
- 00:15 - 00:25 N4010 blue bus ask for right signal

The output video should highlight the vehicle (vehicle_id) and show the event description.

5. Event Export

Save all detected events into a JSON file.


```
{
  "events": [
    {
      "event_type": "touch_line",
      "vehicle_id": "V001",
      "start_time": "00:06",
      "end_time": "00:07"
    },
    {
      "event_type": "change_lane",
      "vehicle_id": "V001",
      "start_time": "00:08",
      "end_time": "00:11"
```

```
    },
    {
      "event_type": "change_lane",
      "vehicle_id": "V002",
      "start_time": "00:14",
      "end_time": "00:17"
    },
    {
      "event_type": "turn_signal",
      "vehicle_id": "V003",
      "signal": "right",
      "start_time": "00:25",
      "end_time": "00:40"
    },
    {
      "event_type": "turn_signal",
      "vehicle_id": "V002",
      "signal": "right",
      "start_time": "00:15",
      "end_time": "00:25"
    }
  ]
}
```


3. **Output Requirements**

UI Expectation

A **video demo** is expected. The minimum UI or demo flow should allow a reviewer to:

- input a video
- run the analysis
- view the lane/line output image

- view the processed event video
- review the generated JSON event output

A simple interface is sufficient. The focus is on functionality, clarity, and explainability.

Test Videos

Some others video from Youtube:

- https://www.youtube.com/watch?v=Y1jTEyb3wiI
- https://www.youtube.com/watch?v=cJatWBDNabE

You may also search for keywords such as:

- 4k traffic detection
- highway traffic 4k
- road traffic surveillance video
- lane change traffic video

Required Deliverables

You must submit:

- a working AI Vision implementation source code
- a formal PDF report
- video demo
- the corresponding JSON event output generated by the system
- lane/line detection image output

Report Expectations

You may use any report structure, as long as it is **logical, clear, and easy to follow**.

Your report must include:

- Architecture diagram
- Flow diagram
- your overall approach
- key design decisions
- implementation overview
- results
- limitations

- evaluation method
- session resource usage such as **CPU / GPU**
- if using a **Vision API**, include token usage and estimated or actual project cost
- if using a **self-hosted / local model**, discuss performance characteristics and trade-offs
- your plan to improve output quality
- scalability considerations, especially if the number of videos, scenes, or traffic conditions grows

You are not required to solve every possible traffic scenario perfectly. A partially completed solution is acceptable if it is well-structured, clearly documented, and demonstrates strong technical reasoning.

End.